

Musharaf Hussain Abid

AI Engineer · LLM Systems, RAG & Computer Vision

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PROFILE

AI Engineer with production experience building and operating LLM-based systems using Python, Azure OpenAI and cloud-native, event-driven architectures. Currently engineering a large-scale document-intelligence and contract-classification pipeline (Azure Service Bus, Azure AI Search, retrieval-augmented generation) for a live enterprise client. Background spans machine learning, deep learning, computer vision and NLP, with a consistent practice of validating system behavior empirically against real production data rather than assumptions.

EXPERIENCE

AI Engineer · Nexora AI (Remote)

June 2026 – Present

- Design and build a production LLM-based contract-intelligence pipeline (Python, Azure OpenAI, Azure Service Bus, Azure AI Search, Azure Blob Storage) that extracts structured data from construction contract documents and classifies inbound project correspondence against standardized contract clause logic, driving automated compliance alerts for a live enterprise client.
- Engineer a hybrid semantic-search correlation system (vector search, BM25, neural reranking) linking related contractual events across a project's full email and document history; iteratively tune matching logic against real production logs rather than fixed heuristics.
- Profiled and optimized a multi-stage document-extraction pipeline, reducing summary-generation latency by 61% (116s to 45s) via parallel task execution; identified and reverted a concurrency increase that degraded throughput due to CPU contention, based on live performance evidence.
- Enforce a deterministic verification layer around all LLM-generated output (dates, monetary values, durations) so financial and temporal accuracy never depends on model-generated text; diagnosed a silent data-loss defect through log-level root-cause analysis.

AI Engineer · Lads Technologies

September 2025 – Present

- Develop and deploy AI-driven solutions using Python and modern AI frameworks for production systems.
- Build and integrate backend APIs supporting scalable application delivery; collaborate cross-functionally on task coordination and code review.

AI Intern · ITSOLERA

April 2024 – October 2024

- Built machine learning pipelines using XGBoost for high-performance structured data modeling; applied BERT-based models for advanced NLP tasks on unstructured text.

Machine Learning Intern · Air University, Islamabad

July 2023 – January 2024

- Implemented a facial-recognition-based attendance system using MTCNN, FaceNet and TensorFlow; automated CSV-based logging with timestamp tracking, eliminating manual attendance recording.

EDUCATION

Air University, Islamabad

Bachelor of Science in Artificial Intelligence. Relevant coursework: Machine Learning, Digital Image Processing, NLP, Deep Learning, Computer Vision, Reinforcement Learning.

TECHNICAL SKILLS

Generative AI & LLM Systems: RAG, LangChain, LlamaIndex, multi-agent / agentic pipelines, prompt engineering, embeddings, LLM output verification

Cloud & Data Platforms: Azure OpenAI, Azure AI Search, Azure Service Bus, Azure Blob Storage, Azure Container Apps, AWS (EC2, S3, SageMaker), Google Vertex AI, Cloudflare Workers, RunPod, Modal

MLOps & Deployment: FastAPI, Streamlit, MLflow, Docker, Git/GitHub, CI/CD pipelines, REST API integration

Languages: Python, C/C++, HTML/CSS

Deep Learning: CNNs, RNNs, LSTMs, Transformers, fine-tuning, computer vision, NLP, TensorFlow, PyTorch

Machine Learning: Data ingestion, preprocessing, feature engineering, model training & evaluation, hyperparameter tuning, XAI, scikit-learn, Pandas, NumPy

Databases & Visualization: MySQL, MongoDB, vector databases (FAISS, Qdrant, Pinecone, Chroma), Matplotlib, Seaborn, Plotly

Tools: Jupyter Notebook, Google Colab, PyCharm, VS Code

SELECTED PROJECTS

Medicine Assistant for Disease Prediction (RoboMedix) · Final Year Project

AI-powered medical assistant predicting diseases and recommending medicine salts from natural patient-bot conversations. Hybrid prediction model fusing XGBoost (structured health data) with a fine-tuned Bio-ClinicalBERT model (clinical text) through a deep learning classifier; Groq-API conversational chatbot for symptom collection; Streamlit interface for real-time inference and explainability.

Stack: TensorFlow, PyTorch, XGBoost, Bio-ClinicalBERT, scikit-learn, Streamlit

PrivateRAG: Secure Enterprise Document Assistant

RAG system enabling employees to securely query internal company documents without exposing confidential data to external APIs; Sentence Transformers embeddings, FAISS/Chroma local vector storage, locally deployed Llama 3 for grounded response generation, Streamlit chat interface.

Stack: Python, LangChain, LLaMA 3, HuggingFace Transformers, FAISS, Chroma

AI-Based Attendance System · Human Activity Recognition · Drowsiness Detection

Facial-recognition attendance (MTCNN detection, FaceNet embeddings, neural classifier, real-time OpenCV capture); CNN+LSTM activity recognition from raw video; real-time driver-drowsiness detection via facial landmark tracking (OpenCV, Dlib).

Stack: TensorFlow, PyTorch, OpenCV, MTCNN, FaceNet, Dlib

CERTIFICATIONS

- Full Stack AI DevOps + Real Projects (AWS, Azure, GCP, K8s) · Udemy
- Artificial Intelligence: ML, Deep Learning, Communication · NAVTTC Prime Minister's Youth Skills Development Program, Government of Pakistan, via Corvit Networks, Rawalpindi
- Operating Systems and You: Becoming a Power User · Google, via Coursera
- Operating System Foundations · Infosec, via Coursera

LANGUAGES

English (professional working proficiency) · Urdu (native) · Punjabi (conversational)

*Full project write-ups, production references and evaluation methodology available on request ·
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